Data Science Fundamentals Assessment

Executive Summary & Objectives

This document serves as a comprehensive assessment covering foundational data science knowledge. It is designed to demonstrate mastery of core concepts including Python programming, statistical analysis, data handling, and industry best practices.

Objectives:

- Demonstrate understanding of core data science concepts, including Python basics, statistical foundations, data types, and industry best practices.
- Complete structured assignments covering all foundational pillars of modern data science.

Module 1: Python Programming for Data Science

Python is the industry-standard language for data manipulation, automation, and modeling. This section evaluates core programming competency.

Key Topics Covered:

• Data Structures: Lists, dictionaries, tuples, and sets.
• Control Flow: Conditional statements, loops, and list comprehensions.
• Functions & Modular Programming: Writing clean, reusable code, lambda functions, and error handling.
• Libraries: Introduction to NumPy for numerical operations and Pandas for tabular data manipulation.

Module 2: Statistical Foundations

Statistics provides the mathematical framework for extracting insights and validating hypotheses from data.

Core Concepts:

• Descriptive Statistics: Mean, median, mode, variance, standard deviation, and percentiles.
• Probability Theory: Probability distributions, conditional probability, and Bayes Theorem.
• Inferential Statistics: Hypothesis testing (Z-test, T-test, Chi-square), p-values, confidence intervals, and A/B testing principles.

Module 3: Data Types and Data Preprocessing

Real-world data is messy. Effective preprocessing ensures data quality and model readiness.

Key Areas:

• Data Types: Numerical (continuous, discrete), categorical (nominal, ordinal), and datetime/text data.
• Handling Missing Data: Imputation techniques vs. removal strategies.
• Feature Engineering: Scaling, normalization, encoding categorical variables (one-hot, label encoding), and outlier detection.

Module 4: Industry Best Practices & Workflow

Applying data science in a professional setting requires adherence to rigorous engineering and ethical standards.

Best Practices:

• Version Control: Utilizing Git and GitHub for tracking code changes and collaboration.
• Reproducibility: Documenting workflows, managing virtual environments (pip/conda), and structuring projects clearly.
• Ethics & Privacy: Data anonymization, compliance with regulations (GDPR/DPDP), and mitigating bias in datasets and machine learning models.

Assessment Deliverables & Conclusion

To successfully complete this assessment, candidates must demonstrate proficiency across all four modules through coding exercises, analytical problem-solving, and structured documentation.